Answer key: Lesson 7, Sharing and Grouping

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Share 8 dots equally among the 4 plates. Draw the dots. Then fill in the blank.

Answer: Each plate has 2 dots

Deal one dot at a time to each of the 4 plates. After 2 rounds all 8 dots are drawn. Each plate has 2 dots, and 4 groups of 2 is 8.

Watch for: Putting all the dots on one plate.; Writing 4, the number of plates.

S2. Mei packs 12 cookies into bags. She puts 3 cookies in each bag. Draw a ring around each group of 3. Then fill in the blank.

Answer: Mei needs 4 bags

12 cookies in groups of 3: ring 3, ring 3, ring 3, ring 3. That is 4 groups, so Mei needs 4 bags. Count in jumps of 3 to check: 3, 6, 9, 12.

Watch for: Writing 3, the number in each bag.; Ringing groups of different sizes.

S3. Ali has 14 sweets. He shares them equally among 2 friends. How many sweets does each friend get?

Answer: Each friend gets 7 sweets

Dealing 14 sweets to 2 friends one at a time gives 7 each. Check: 2 groups of 7 is 14.

Watch for: Writing 2, the number of friends.; Writing 14 or 16 by adding.

S4. Share the sweets equally among the friends. Fill in the table.

Answer: 4, 3, 4

8 sweets among 2 friends is 4 each. 12 sweets among 4 friends is 3 each. 20 sweets among 5 friends is 4 each. Check: 2 groups of 4 is 8, 4 groups of 3 is 12, 5 groups of 4 is 20.

Watch for: Writing the number of friends in the last column.; Giving 5 for the last row because it has a 5.

S5. Ben puts 20 pencils into pots. He puts 5 pencils in each pot. How many pots does Ben use?

Answer: Ben uses 4 pots

5, 10, 15, 20 is 4 jumps, so Ben uses 4 pots. Check: 4 groups of 5 is 20.

Watch for: Writing 5, the number in each pot.; Stopping at 15.

S6. Mei shares 12 stickers equally among 4 friends. Each friend gets 3 stickers.

Answer: 4 × 3 = 12 (3 × 4 = 12 is also right); Yes

4 friends with 3 stickers each is 4 groups of 3: 4 × 3 = 12. (3 × 4 = 12 is also right.) That is the 12 stickers Mei began with, so the answer matches.

Watch for: Writing 4 × 3 = 7 by adding.; Circling No because the numbers look different.

S7. Read each story. Circle Share or Group.

Answer: Share; Group

Lily deals the beads out equally to 3 friends, so it is Share (4 beads each). Jamie puts 6 beads in each box, so it is Group (2 boxes).

Watch for: Choosing Share because both stories have 12 beads.; Choosing by the number 3 or 6.

Think harder (page 3)

T1. Ali has 19 counters. He puts them into groups of 4.

Answer: 4 groups; 3 left over

4, 8, 12, 16 is 4 groups, and 16 counters are used. 19 - 16 = 3 counters are left. Three is not enough for another group of 4.

Watch for: Writing 5 groups by rounding up.; Forgetting the counters left over.

T2. Ben shares 20 biscuits equally with his 3 cousins. Aisha says 3 people share the biscuits.

Answer: No; 5 biscuits each

Ben shares with his 3 cousins, so Ben is one of the sharers: Ben and 3 cousins is 4 people, not 3. 20 biscuits among 4 people is 5 each, because 4 groups of 5 is 20. (A story that says 'shares among 3 friends' does not count the sharer: there the 3 friends are all the people.)

Watch for: Saying Yes because the story says 3 cousins.; Sharing 20 among 3 people.

T3. Mei has 18 sweets. Read the two stories.

Answer: Group; 3 bags; Share; 3 sweets

Story A tells us 6 sweets in each bag, so it is Group, and 18 in groups of 6 makes 3 bags. Story B tells us 6 bags, so it is Share, and 18 shared among 6 bags is 3 in each bag. Both answers are 3, but the meanings are different.

Watch for: Saying Share for both stories.; Answering 6 for (b) or (d).

T4. Ravi has 17 stickers. He puts 5 stickers in each bag.

Answer: 3 bags; 2 left over; 3 more

3 bags use 15 stickers, so 2 stickers are left over. One more bag needs 5 stickers, and 2 + 3 = 5, so he needs 3 more.

Watch for: Writing 2 for (c), the number left over.; Writing 4 bags.

Go further (page 4)

G1. Jenny has 12 sweets. She shares them equally among some friends. Fill in the table. Then answer (a) and (b).

Answer: 12, 6, 4, 3, 2, 1; No; 2 left over; 18 sweets: 1, 2, 3, 6, 9, 18 friends

1 friend gets all 12. 2 friends get 6 each. 3 friends get 4 each. 4 friends get 3 each. 6 friends get 2 each. 12 friends get 1 each. With 5 friends, each gets 2 and that uses 10 sweets, so 2 are left over and the answer to (a) is No. (c) With 18 sweets: 1 friend gets 18, 2 get 9, 3 get 6, 6 get 3, 9 get 2 and 18 get 1. Try 4, 5, 7 and 8: each leaves some over. So the numbers are 1, 2, 3, 6, 9 and 18, six ways in all. Finding every way to share is the start of factors, which school meets later.

Watch for: Writing the number of friends instead of the number each.; Saying Yes for 5 friends.; Part (c): stopping at 2, 3 and 6 and missing 9 or 18, or writing how many each gets instead of the number of friends.

G2. Jenny shares 20 pencils equally among 4 pots. Then she puts 3 more pencils in each pot.

Answer: 5 pencils; 8 pencils; 32 pencils

20 pencils among 4 pots is 5 in each pot. Then 3 more in each pot makes 5 + 3 = 8 in each pot. 4 pots with 8 pencils is 8 + 8 + 8 + 8 = 32 pencils in all.

Watch for: Adding 3 to 20.; Stopping after 5.

G3. Wei has 20 stickers. He keeps 2 stickers. He shares the rest equally among 3 friends.

Answer: 18 stickers; 6 stickers each

20 - 2 = 18 stickers are shared. 18 shared among 3 friends is 6 each, because 3 groups of 6 is 18.

Watch for: Sharing all 20 stickers.; Writing 7 by sharing 20 among 3.

Investigation (page 5): Sharing among 4 plates

Left over: 1, 2, 3, 0, 1, 2, 3, 0. The left-over numbers 0, 1, 2, 3 repeat. Never more than 3 are left over, because with 4 left over we could give one more counter to every plate.

Look for: A child who deals out the counters for several numbers, writes the left-over numbers and notices that they repeat 1, 2, 3, 0. Credit a child who says the same numbers come back every 4 counters, or who says the left over is never as big as the number of plates. Do not mark down a child who finishes only some of the numbers.

Exit ticket (page 5)

E1. 20 sweets are shared equally among 4 children. Fill in the blank.

Answer: 5

Deal 20 sweets to 4 children one at a time: 5 each. Check: 4 groups of 5 is 20.

Watch for: Writing 4, the number of children.; Writing 24 by adding.

E2. Which story puts 12 sweets into groups of 4? Write its number in the bracket.

Answer: (2) Mei puts 12 sweets into bags of 4.

In story 2 the sweets are put into bags of 4, so they are in groups of 4 and there are 3 bags. Story 1 shares among 4 friends (3 each). Stories 3 and 4 do not make groups.

Watch for: Choosing story 1 because it has a 4 in it.; Choosing story 3 because it takes away 4.

E3. Ali puts 18 cards into piles. Each pile has 3 cards.

Answer: 6 piles; 6 × 3 = 18

Count in jumps of 3 to 18: 3, 6, 9, 12, 15, 18 is 6 jumps, so 6 piles. Check: 6 × 3 = 18.

Watch for: Writing 3, the number in each pile.; Writing 15 by stopping early.

Sneak-peek puzzles (not in the workbook)

1. 3
2. Mei. Ali puts 3 sweets in each bag (4 bags). Mei shares 12 among 3 bags, so each bag has 4.